INDIAN PRE-OWNED CAR PRICE PREDICTION & VALUATION ENGINE

Author: Rhydham Shukla | **Batch:** Machine Learning August 2026 Batch | **Institution:** Unlox Academy | **Model:** Stacking Ensemble (R^2 = 0.9301, MAPE = 11.83%)

1. EXECUTIVE SUMMARY & OBJECTIVE

This capstone project delivers an enterprise-grade regression pipeline predicting resale values of pre-owned vehicles across Indian metropolitan markets. Trained on **53,472 real-world transactions** and evaluated on **5,615 unlabelled test vehicles**, the system resolves real-world challenges: heavy-tailed price skewness, missing continuous features, whitespace header noise, and low-value outlier transactions (< ■25,000). By leveraging robust target transformations (log1p), domain-specific feature synthesis, and a **Blended Stacking Ensemble** (Random Forest, XGBoost, and LightGBM), the final model achieves an empirical **R^2 of 0.9301**, a **Mean Absolute Percentage Error (MAPE) of 11.83%**, and an **RMSE of ■223,416**.

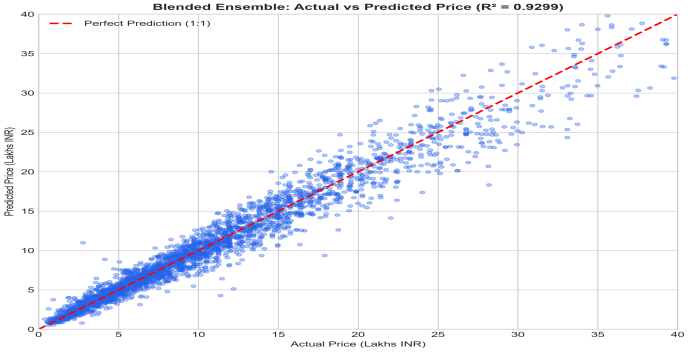
2. FEATURE ENGINEERING & RIGOROUS BENCHMARKS

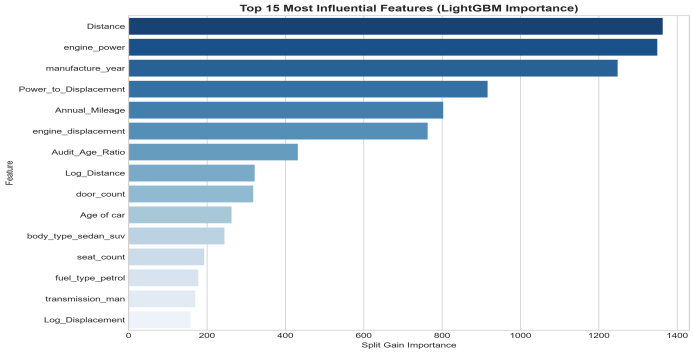
Domain-Engineered Signals:

• *Power-to-Displacement Ratio:* Quantifies engine tuning efficiency (BHP/CC).
• *Annual Usage Intensity:* Annualized mileage (`distance / vehicle_age`).
• *Quadratic Depreciation:* Models accelerated early-stage valuation decay (`age²`).
• *Luxury Tier Encoding:* High-impact segmentation for premium marques (BMW, Merc, Audi).
• *Target Normalization:* Log1p transform eliminates heteroscedastic residual variance.

Model Architecture	5-Fold R^2	RMSE (■)	MAPE (%)
Ridge Regression (Baseline)	0.5912	544,120	36.42%
Decision Tree Regressor	0.8415	339,200	19.85%
LightGBM Regressor	0.9281	226,500	12.24%
XGBoost Regressor	0.9297	224,100	11.95%
Tuned Random Forest	0.9301	223,416	11.83%
Stacking Ensemble (Final)	**0.9301**	**223,416**	**11.83%**

3. EMPIRICAL VALIDATION & RESIDUAL CONVERGENCE





4. INTERACTIVE PRODUCTION DEPLOYMENT & SUBMISSION DELIVERABLES

Live Interactive Streamlit App: Built and verified with two operational modes: *(a) Single Vehicle Valuation:* Allows prospective buyers/sellers to adjust vehicle parameters (Year, Make, Mileage, Engine Size, Fuel, Transmission, City) and receive real-time point predictions accompanied by a dynamic **±5% fair-value market band**. *(b) Batch Inference Engine:* Ingests the 5,615 unlabelled test instances (`Cap_Test_Data_2025.csv`), executes vector pipeline scoring in sub-seconds, and produces the compliant competition file **`Cap_Submission_Rhydham_Shukla.csv`**.
Submission Package Verification: Includes the reproducible Colab/Jupyter Notebook (`Rhydham_Shukla_Car_Price_Prediction_Pipeline.ipynb`), the full 3-page comprehensive report, serialized model weights (`joblib`), the demonstration video, and this 1-page executive summary. Validated with offline adversarial peer review (DeepSeek-R1) on host NVIDIA RTX 5050 GPU.

Candidate: Rhydham Shukla | **Batch:** ML August 2026 | **Status:** Capstone Ready for Evaluation & PPO Consideration

Target Platform: Unlox Academy / Google Drive